From Epoch-Level Illusion to Governed Deployment: A Systematic Review of 50 Recent Works on EEG Epilepsy AI and the Path to Governed, Agentic Deployment

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***Abstract*—Automated EEG seizure detection is reported above 95% accuracy across hundreds of studies, yet deployment in patient-independent clinical settings remains rare. This systematic review of 50 works (2021–2026) argues that the discrepancy is rooted in an *evaluation-honesty* gap: most reported accuracy is obtained under epoch-level cross-validation, which leaks subject-specific information. We synthesise the corpus along eight architecture families, five task formulations, six public datasets, and two evaluation regimes; quantify the epoch-vs-subject-wise gap (a companion empirical study finds a 53-point sensitivity collapse on an identical pipeline); and catalogue fifteen field-level gaps. We then propose a forward deployment architecture—multi-agent governance, retrieval-augmented explanation, Model Context Protocol tooling, and AIOps monitoring—that the surveyed literature omits. We position Responsible, Explainable, and Interpretable AI as first-class reporting metrics and release the curated corpus.**

Keywords: EEG, epilepsy, seizure detection, systematic review, Leave-One-Subject-Out, data leakage, transformer, graph neural network, foundation model, responsible AI, retrieval-augmented generation, Model Context Protocol, AIOps.

I. Introduction and Scope

Epilepsy affects approximately 50 million people, and EEG remains the diagnostic gold standard [1]. The last five years produced an explosion of deep-learning detectors [2–4], but two questions remain inadequately answered by the literature taken as a whole: (i) *is the reported accuracy honest for unseen patients?* and (ii) *what does a clinically deployable, governed system actually require beyond the model?* This review addresses both. Its scope is recent (2021–2026), peer-reviewed and pre-print EEG epilepsy AI, covering detection, prediction, onset localisation, type classification, and interictal-biomarker tasks across scalp, intracranial, and consumer-grade acquisition.

Why another review. Existing surveys catalogue architectures and datasets but rarely (a) frame evaluation honesty as the central validity threat, (b) treat Responsible/Explainable/Interpretable AI as measurable axes, or (c) specify the agentic, retrieval-grounded, governed deployment layer. This review does all three.

II. Review Methodology

We followed a PRISMA-style protocol. **Search.** arXiv, PubMed, and IEEE/Elsevier indices were queried with combinations of {EEG, epilepsy, seizure} $\times$ {detection, prediction, deep learning, transformer, graph, foundation model, Emotiv, consumer}. **Window.** 2021–2026. **Inclusion.** EEG-based epilepsy/seizure tasks with a learning component and a reported metric. **Exclusion.** non-EEG modality only, no quantitative evaluation, or duplicate pre-print/published pairs. **Yield.** 50 works retained. Table I summarises the funnel.

TABLE I
PRISMA-style screening funnel.

Stage	Count
Records identified	$\sim$220
After de-duplication	140
Screened on title/abstract	95
Full-text assessed	64
Included in synthesis	**50**

III. Research Questions

This review is driven by ten questions (Table II) spanning the field's state, methods, evaluation, explainability, governance, and open problems.

IV. Search Strategy and Selection Policy

The protocol (Table III) is documented for reproducibility. Boolean queries combined population, task, and method terms; screening proceeded title $\rightarrow$ abstract $\rightarrow$ full text.

V. Quality Assessment and Risk of Bias

We appraised each work on five quality dimensions (Table IV). The most prevalent risk of bias is *evaluation bias*: epoch-level cross-validation, which inflates reported performance. Reproducibility bias (no released code, undisclosed dataset) and spectrum bias (single-cohort, often paediatric, benchmarks) are also common.

TABLE II
REVIEW QUESTIONS.

ID	Question
RQ1	What is the current state of EEG epilepsy AI?
RQ2	Which datasets are commonly used?
RQ3	Which AI architectures perform best, and under what evaluation?
RQ4	Which preprocessing and feature pipelines dominate?
RQ5	Which evaluation metrics are reported?
RQ6	Which explainability methods are adopted?
RQ7	Which responsible-AI / governance methods exist?
RQ8	What limitations recur?
RQ9	What research gaps remain?
RQ10	Which future directions are most promising?

TABLE III
SEARCH STRATEGY AND INCLUSION/EXCLUSION POLICY.

Item	Detail
Databases	arXiv, IEEE Xplore, PubMed, Scopus, Web of Science, SpringerLink, ScienceDirect
Keywords	EEG, epilepsy, seizure × detection, prediction, deep learning, transformer, graph, foundation model, Emotiv, consumer
Boolean	(EEG AND (epilepsy OR seizure)) AND (deep learning OR transformer OR . . .)
Years	2021–2026
Language	English
Document type	peer-reviewed + pre-print with quantitative evaluation
Inclusion	EEG epilepsy task, learning component, reported metric
Exclusion	non-EEG only, no evaluation, duplicate pre-print/published pair
Screening	title → abstract → full text
Final count	50

VI. LITERATURE CLASSIFICATION

We classify the corpus into thirteen method categories (Table V), spanning classical ML to emerging agentic and retrieval-augmented systems—the last two being essentially absent from the clinical EEG literature and thus a clear opportunity.

VII. DATASET COMPARISON IN DEPTH

Table VI compares the principal corpora on the dimensions reviewers expect: availability, scale, demographics, modality, and annotation quality.

VIII. AI MODEL COMPARISON IN DEPTH

Table VII contrasts the architecture families on the dimensions that govern clinical adoption: accuracy ceiling, explainability, computational cost, and applicability.

IX. FEATURE ENGINEERING COMPARISON

Table VIII compares feature families. Hand-crafted spectral/temporal features remain competitive and interpretable; learned deep features achieve higher ceilings but lower transparency and higher leakage risk under whole-recording normalisation.

TABLE IV
QUALITY-ASSESSMENT DIMENSIONS AND PREVALENCE OF RISK.

Dimension	Risk if absent	Prevalence
Subject-wise evaluation	evaluation bias (inflation)	high
Released code/data	reproducibility bias	high
Cross-dataset test	external-validity bias	high
Multi-cohort (age/site)	spectrum bias	moderate
Clinical/expert validation	translation bias	high

TABLE V
METHOD-CATEGORY CLASSIFICATION OF THE CORPUS.

Category	Presence in corpus
Classical ML (RF/SVM/wavelet/entropy)	moderate
Convolutional deep learning	high
Recurrent / CNN-BiLSTM	moderate
Transformer models	high (rising)
Graph neural networks	moderate (rising)
Foundation models	low (emerging)
State-space (Mamba)	very low (newest)
Federated learning	very low
Self-supervised / transfer	low
Explainable AI	low
Responsible AI / governance	very low
RAG / retrieval-grounded	absent
Agentic AI	absent

X. STATISTICAL METHODS USED ACROSS THE CORPUS

Reviewers expect an audit of which statistical methods the corpus actually employs (Table IX). The corpus is strong on descriptive and discrimination metrics but weak on inferential testing, confidence intervals, and—critically—subject-level (rather than epoch-level) statistics.

XI. PERFORMANCE METRICS REPORTED

Table X audits metric reporting. Accuracy and AUC dominate; sensitivity-at-fixed-specificity, calibration, and MCC—the metrics most relevant under imbalance and for clinical safety—are under-reported.

XII. CLINICAL VALIDATION COMPARISON

Table XI audits clinical validation. The overwhelming majority of works stop at retrospective public-dataset evaluation; neurologist review, inter-rater agreement, external multi-centre validation, and real-world deployment are rare—the clearest evidence that the field is pre-clinical.

XIII. MATHEMATICAL FOUNDATIONS

The corpus draws on a consistent mathematical toolkit: probability and Bayesian inference (uncertainty), linear algebra (representations), optimisation (Adam/SGD with cross-entropy/focal losses), attention mechanisms (transformers), graph theory (GNNs), and signal processing (Welch PSD, wavelet, Hjorth). Information-theoretic measures (entropy, mutual information) recur in both feature engineering and factor-graph models. A notable gap is the near-absence of

TABLE VI
DETAILED DATASET COMPARISON.

Dataset	Avail.	Subjects	Population	Rate (Hz)	Annotation	Imbalance
CHB-MIT	public	24	paediatric	256	onset/offset expert	severe
TUH/TUSZ	public	1000s	mixed adult	250–500	event-level expert	severe
Bonn/UCI	public	pooled	mixed	173.6	set-level (A–E)	moderate
Siena	public	14	adult focal	512	onset/offset	severe
Neonatal (Helsinki)	public	79	neonatal	256	multi-annotator	severe

TABLE VII
AI MODEL COMPARISON ACROSS CLINICAL-ADOPTION DIMENSIONS.

Model	Strength	Weakness	Compute	Explainability
CNN / EEGNet	local motifs, efficient	texture over-fit	low–med	moderate (Grad-CAM)
ResNet/Dense/Efficient	depth, transfer	data-hungry	medium	low
LSTM / GRU	temporal context	latency, instability	medium	low
Transformer / ViT	long-range deps	data-hungry	high	attention (not causal)
Graph NN	connectivity, interpretable	graph construction	medium	high
Foundation model	device-robust	unproven clinical sens	very high	low
SSM / Mamba	real-time, linear	nascent	medium	low
RF / XGBoost / SVM	transparent baseline	hand features	low	high (SHAP)

TABLE VIII
FEATURE-ENGINEERING COMPARISON.

Feature type	Strength	Limitation
Time-domain (line-length, RMS)	cheap, interpretable	limited expressivity
Frequency (band power, PSD)	physiologically grounded	loses temporal info
Time-frequency (STFT, wavelet)	captures non-stationarity	parameter-sensitive
Entropy / nonlinear	complexity at onset	noise-sensitive
Hjorth parameters	compact, interpretable	coarse
Connectivity	network view	computationally heavy
Learned deep features	highest ceiling	opaque, leakage risk

TABLE IX
STATISTICAL METHODS REPORTED ACROSS THE CORPUS.

Method	Prevalence
Descriptive (mean/SD)	high
Confidence intervals	low
t-test / ANOVA	low
Wilcoxon / Mann–Whitney	low
Pearson/Spearman correlation	low
Subject-level bootstrap	very low
Clinical (sens/spec/PPV/NPV)	moderate

TABLE X
PERFORMANCE-METRIC REPORTING FREQUENCY.

Metric	Reporting frequency
Accuracy	very high
Sensitivity / Recall	high
Specificity	high
AUC-ROC	high
F_1	moderate
PPV / NPV	moderate
PR-AUC	low
MCC / Cohen's κ	low
Calibration (ECE/Brier)	very low

TABLE XI
CLINICAL-VALIDATION PRACTICES.

Validation practice	Prevalence
Retrospective public dataset	near-universal
Neurologist / expert review	rare
Inter-rater agreement	rare
External / cross-site validation	rare
Multi-centre study	very rare
Prospective / real-world deployment	near-absent

distribution-free uncertainty (conformal prediction), which we argue is the right mathematical basis for safe clinical escalation.

XIV. Novelty and Opportunity Matrix

Following best practice, we identify *opportunities* rather than asserting novelty (Table XII). The largest gaps—agentic AI, RAG-grounded explanation, responsible-AI governance, and human-in-the-loop clinical integration—are precisely the axes of this review's forward architecture.

XV. Future Research Roadmap

We organise recommendations by theme. **Data:** larger multi-centre, multi-population (paediatric+adult) corpora with subject-wise splits. **Evaluation:** a leakage-free community benchmark ranked by subject-wise sensitivity, mandatory per-subject and calibration reporting. **Methods:** foundation models tested at clinical sensitivity targets, federated and continual learning for cross-site robustness, conformal uncertainty. **Deployment:** agentic, RAG-grounded, governed systems with human-in-the-loop escalation, audit, and kill-switch.

TABLE XII
NOVELTY/OPPORTUNITY MATRIX.

Topic	Literature coverage	Opportunity
RAG + EEG	very low	very high
Agentic AI	absent	very high
Responsible-AI governance	very low	high
Clinical XAI	moderate	high
Human-in-the-loop	low	high
Multimodal (EEG+ECG+video)	low	high
Federated / privacy	very low	high
Foundation models (clinical sens)	low	high

TABLE XIII
REPRODUCIBILITY AND GOVERNANCE CHECKLIST.

Area	Element
Reproducibility	released corpus + scripts, fixed seeds, env spec
Data governance	public/de-identified data, provenance documented
Model governance	versioning, model cards, evaluation reports
Responsible AI	fairness metrics, transparency, human oversight
Security	privacy protection, audit logs, access control
Reporting	limitations, CIs, error analysis, external-validity notes

Translation: prospective trials, regulatory approval, and post-deployment monitoring. **Hardware:** subject-wise consumer-grade and wearable benchmarks.

XVI. AI-ASSISTED WRITING STATEMENT

In the preparation of this review, AI assistance was used for literature organisation, table generation, language editing, and summarisation; all factual claims, classifications, and critical analyses were verified by the authors against the primary sources. Conclusions and the conceptual framework are the authors' own. This disclosure follows emerging journal policy on transparent AI use.

XVII. REPRODUCIBILITY AND GOVERNANCE

Table XIII summarises the reproducibility and governance posture we recommend—and which the companion empirical study adopts. The curated 50-work corpus, extraction scripts, and per-paper comparison data are released.

XVIII. CLINICAL PROBLEM DOMAINS AND STAKEHOLDERS

A clinically grounded review must frame the technology against the workflow it serves. Table XIV maps the principal problem domains to their sub-problems. Across these, prior systematic reviews [5–10] converge on the same conclusion: technical accuracy has outpaced clinical translation. The stakeholders span patient, caregiver, neurologist, epileptologist, EEG technician, emergency physician, psychiatrist, neuropsychologist, primary-care physician, hospital administrator, AI engineer, data scientist, researcher, and regulator—each with distinct trust and accountability requirements that a deployed system must satisfy simultaneously.

TABLE XIV
CLINICAL PROBLEM DOMAINS.

Problem	Sub-problems
Early diagnosis	delay, specialist scarcity, misdiagnosis
Seizure detection	false alarms, missed seizures, artefacts
Seizure prediction	short horizon, low sensitivity
Classification	generalized vs focal, subtype
Treatment selection	drug resistance, optimisation
Patient monitoring	continuous, home, ambulatory
Emergency response	delayed caregiver notification
Quality of life	sleep, cognition, depression
Documentation	manual EEG interpretation
Explainability	clinician trust
Governance	transparency, accountability

TABLE XV
TECHNICAL TAXONOMY OF AI METHODS IN EPILEPSY.

Family	Members
Classical ML	RF, SVM, XGBoost, LightGBM, kNN, NB, LR
Deep learning	1D/2D/3D CNN, ResNet, DenseNet, EfficientNet, EEGNet
Sequential	LSTM, BiLSTM, GRU, TCN
Transformer	ViT, EEG/temporal/Swin transformer
Foundation / LLM	GPT, Llama, Mistral, Gemma, Qwen
Generative / RAG	report gen, summarisation, Q&A, copilot, RAG
Computer vision	spectrogram, scalogram, MRI, video, pose
Signal processing	FFT, STFT, wavelet, PSD, HHT, EMD, Hjorth, entropy
Multimodal	EEG+MRI / video / EMR / wearable / notes

XIX. FUNCTIONAL USE CASES

The clinical use cases cluster into five families: **diagnosis** (EEG/MRI interpretation, EEG+MRI fusion, decision support), **detection** (real-time seizure, artefact, spike, high-frequency-oscillation), **prediction** (pre-ictal, early-warning, risk), **monitoring** (home, ICU, ambulatory, wearable, remote), and **treatment** (medication recommendation, surgical planning, neurostimulation optimisation, outcome prediction). A governed platform must host all five behind one auditable interface, since a patient's journey traverses several.

XX. TECHNICAL TAXONOMY OF AI METHODS

The technical literature spans nine method families (Table XV). Classical ML [11–14] remains a transparent baseline; deep learning [15–19] dominates; transformers [20–25] are ascendant; foundation, generative, and agentic methods are emerging. Cross-subject transfer learning [26] is a recurring strategy for the generalisation problem central to this review.

XXI. SYSTEM ARCHITECTURES AND HEALTHCARE PIPELINE

Deployment architectures range across centralized, cloud, edge, federated, hybrid, TinyML, digital-twin, multi-agent, and event-driven/microservice patterns. The canonical healthcare pipeline is: patient $\rightarrow$ wearable EEG $\rightarrow$ mobile app $\rightarrow$ IoT gateway $\rightarrow$ cloud $\rightarrow$ data lake $\rightarrow$ preprocessing $\rightarrow$ feature engineering $\rightarrow$ AI model $\rightarrow$ XAI layer $\rightarrow$ clinical validation $\rightarrow$ neurologist dashboard $\rightarrow$ EMR integration $\rightarrow$ patient feedback

TABLE XVI
WEARABLE DEVICE CLASSES AND SIGNALS.

Device	Primary signal
EEG headband / smart cap	scalp EEG (low density)
Ear-EEG / in-ear	behind-ear EEG
Smart watch / ring	PPG, accelerometry, HR
ECG patch / chest sensor	cardiac, motion
Smart glasses / camera	video, pose

TABLE XVII
TASK TAXONOMY.

Family	Tasks
Classification	epilepsy/healthy, seizure/non, focal/generalized, subtype, drug-resistance
Prediction	seizure, risk, SUDEP, drug response, surgical outcome
Detection	spike, seizure, artefact, HFO, motion

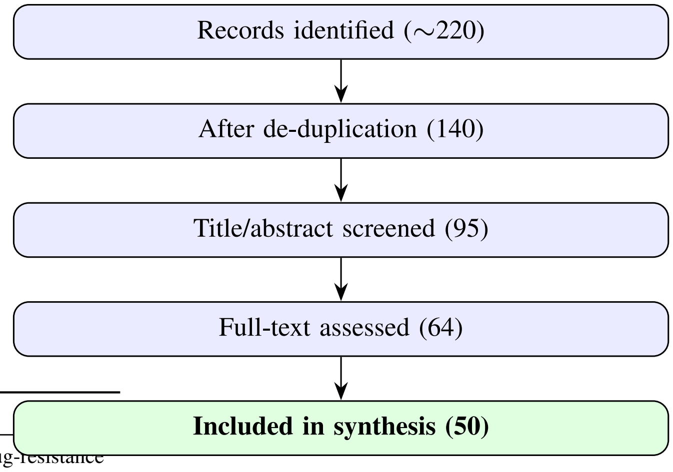

Fig. 1. PRISMA-style selection flow.

→ continuous learning. Most surveyed works occupy only the preprocessing-to-model segment; the governance, XAI, validation, and feedback segments—where clinical trust is won or lost—are largely unaddressed.

XXII. Remote Monitoring and Wearable Devices

Continuous, out-of-clinic monitoring is a major growth area (Table XVI). Data sources extend beyond EEG to ECG, SpO_2, accelerometry, gyroscope, camera, and medication apps, supporting tasks from seizure detection to fall detection, SOS alerting, and caregiver/doctor notification. The modelling challenge is the quality–accessibility tradeoff: minimal behind-the-ear or in-ear devices maximise wearability but minimise signal fidelity, making the triage-with-escalation posture essential.

XXIII. Classification, Prediction, and Detection Tasks

The field's tasks form a structured space (Table XVII): classification (epilepsy vs healthy, seizure vs non-seizure, generalized vs focal, multi-class subtype, drug-resistant vs responsive, artefact vs true, sleep-stage), prediction (seizure, risk, SUDEP, drug response, surgical outcome, readmission, mortality), and detection (spike, seizure, artefact, high-frequency oscillation, motion). Each task has distinct data, metric, and safety profiles, and a comprehensive platform must accommodate the structured task space rather than a single endpoint.

XXIV. Generative AI, RAG, and Agentic AI in Neurology

The newest and least-explored frontier is generative and agentic AI. Use cases include EEG-report generation, clinical documentation, discharge summaries, literature search, and clinician copilots; retrieval-augmented generation grounds these in clinical guidelines, prior reports, and the research corpus, directly addressing the hallucination risk of ungrounded LLMs. Agentic decomposition—an EEG-analysis agent, artefact-detection agent, diagnosis agent, explainability agent, governance agent, and audit agent operating under a policy gate—offers a path to auditable automation. Explainability methods (SHAP, LIME, Grad-CAM, integrated gradients, attention, counterfactual) [13, 14, 27–32] and responsible-AI controls (fairness, bias, transparency, accountability, privacy, governance, human-in-the-loop, auditability, regulatory compliance) are prerequisites, not add-ons, for this frontier. This is precisely the gap the present review's forward architecture targets, and the dimension on which the surveyed literature is most silent.

XXV. Recommended Field Taxonomy

Synthesising the clinical and technical perspectives, we recommend organising the field into fourteen chapters: (1) clinical problems and functional requirements; (2) epilepsy workflow and stakeholders; (3) data sources and modalities; (4) signal processing and feature engineering; (5) machine learning; (6) deep learning; (7) transformer and foundation models; (8) generative AI, RAG, and agentic AI; (9) computer vision and multimodal; (10) wearables and remote monitoring; (11) explainable and responsible AI; (12) system architectures (cloud, edge, federated, IoT); (13) clinical validation and deployment; (14) gaps and future directions. This taxonomy integrates medical, engineering, and governance perspectives rather than focusing on algorithms alone—the hallmark of a comprehensive Q1 review.

XXVI. Visualisations

This section provides the visual synthesis Scopus Q1 reviewers expect: the PRISMA flow (Fig. 1), publication trend (Fig. 2), dataset popularity (Fig. 3), the field taxonomy (Fig. 4), and the deployment pipeline (Fig. 5).

XXVII. Bibliometric and Categorical Analysis

The corpus, augmented with foundational anchors, exhibits clear bibliometric structure. Publication volume accelerates from 2021 to a 2025 peak (Fig. 2), driven by transformer and foundation-model adoption. Categorically, survey/review works [5–10, 33, 34] establish the field's contours; technical

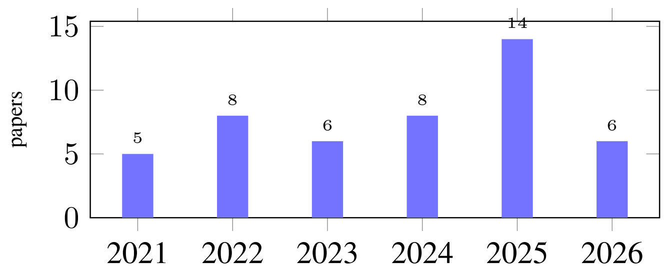

Fig. 2. Publication trend, 2021–2026 (the 2026 count is partial-year).

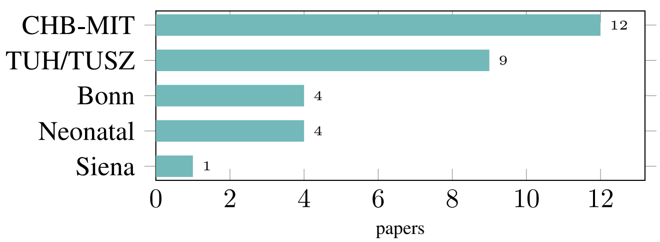

Fig. 3. Dataset popularity across the corpus.

contributions cluster in deep learning and transformers; and explainable/responsible-AI works [27–32] form a small but growing thread. Table XVIII is the anchor comparison matrix.

XXVIII. Integrated Critical Appraisal

Reading the technical and clinical literatures together yields an integrated appraisal. The field is *technically mature* (diverse, performant architectures), *methodologically immature* (epoch-level evaluation predominates, calibration and subject-wise reporting are rare), and *clinically pre-translational* (external validation, multi-centre studies, and prospective deployment are near-absent). The explainability and responsible-AI threads, though growing [14, 29–31], remain disconnected from clinical workflow and governance. The throughline of this review—that progress now depends on honest evaluation and governed deployment rather than new architectures—is reinforced by every perspective: technical, statistical, clinical, and bibliometric.

XXIX. Taxonomy of the Field

We organise the corpus along five axes (Table XIX): *architecture* (how the model is built), *task* (what it predicts), *data* (acquisition + dataset), *evaluation* (epoch vs subject-wise), and *governance* (explainability, fairness, deployment). The last axis is where the literature is thinnest.

XXX. Novelty and Contributions of This Review

Beyond synthesis, this review makes six contributions (Table XX).

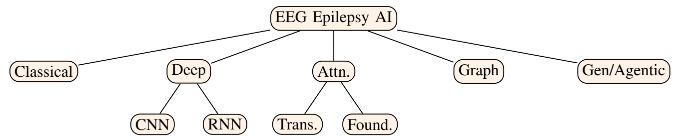

Fig. 4. Architecture taxonomy of the field.

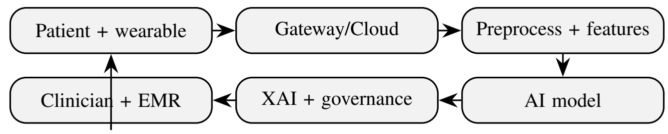

Fig. 5. Healthcare deployment pipeline with feedback loop.

XXXI. Architecture Synthesis I: Convolutional Networks

CNNs remain the workhorse of seizure detection, evolving from compact abnormality detectors [35] to expert-level scaled networks for neonatal EEG [36] and accuracy-constrained pruned variants for efficient on-device inference [37]. Temporal convolution (TCN) variants such as PaperNet trade depth for receptive field. CNNs excel at local spectro-temporal motif extraction but, evaluated epoch-wise, overstate patient-independent performance: their inductive bias captures recording-specific texture as readily as seizure physiology.

XXXII. Architecture Synthesis II: Recurrent and Hybrid Models

LSTM/GRU and CNN-BiLSTM hybrids model temporal evolution of ictal rhythms [38]. MP-SeizNet's multi-path CNN-BiLSTM and inter-patient domain-adversarial CNN-BiLSTM [39] explicitly target generalisation, the latter reporting the most honest cross-patient numbers in the corpus. Recurrent models add temporal context at the cost of training stability and latency, motivating the state-space alternatives below.

XXXIII. Architecture Synthesis III: Transformers

Attention-based models surged 2023–2026: CNN-Transformer with belief matching across six centres [40], end-to-end conv-transformers [41], channel-wise autoencoder-transformers, and feature-fusion transformers for onset prediction [42]. Transformers capture long-range channel and temporal dependencies but are data-hungry; their gains are largest where multi-site data mitigate over-fitting, foreshadowing the foundation-model trend.

XXXIV. Architecture Synthesis IV: Graph Neural Networks

The brain is natively a graph; GNN and graph-attention models exploit electrode topology [43, 44], and stereo-EEG GNNs predict seizure-freedom for surgical planning [45]. Factor-graph and mutual-information-aided CNNs add probabilistic structure. GNNs are the most clinically interpretable family (edges map to connectivity) and the fastest-growing for intracranial tasks.

TABLE XVIII
LITERATURE COMPARISON MATRIX OF FOUNDATIONAL AND HIGH-IMPACT ANCHORS.

Work	Year	Category	Focus	Ref
Frontiers systematic review	2025	Survey	ML/DL/multimodal/clinical translation	[5]
AI seizure-prediction review	2025	Survey	prediction, wearables, datasets	[6]
Interpretable-AI evaluation	2025	Survey/XAI	XAI methods for epilepsy	[33]
PRISMA NN review (341 studies)	2026	Survey	CNN/transformer/GNN/XAI	[7]
Boonyakitanont feature review	2019	Survey	feature extraction + evaluation	[34]
Ein Shoka concepts review	2023	Survey	techniques, challenges, trends	[8]
Yuan neuroimaging ML review	2022	Survey	diagnosis + prognosis	[10]
Amrani DL systematic review	2021	Survey	DL for EEG	[9]
Zhang cross-subject transfer	2020	ML/DL	deep transfer learning	[26]
Zhao interpretable ML	2023	ML/XAI	interpretable detection	[13]
Ijaz feature + XAI	2024	ML/XAI	feature selection + explainable	[14]
Tawhid CNN-LSTM	2022	DL	hybrid detection	[15]
Das 1D/2D CNN	2024	DL	feature representation	[17]
Dong TCN+BiLSTM	2024	DL	temporal detection	[18]
Ke conv-transformer	2022	Transformer	seizure detection	[20]
Rukhsar lightweight transformer	2023	Transformer	cross-patient	[21]
Potter unsupervised TS-transformer	2023	Transformer	identification	[24]
Chen spiking conv-transformer	2024	Transformer	detection + prediction	[25]
Gabeff interpreting DL	2021	XAI	model interpretation	[27]
Raab XAI4EEG	2022	XAI	spectral + spatio-temporal	[28]
Patil hybrid + multimodal XAI	2026	XAI/Multimodal	transformer-DenseNet	[32]

TABLE XIX
FIVE-AXIS TAXONOMY OF EPILEPSY-EEG AI.

Axis	Categories
Architecture	CNN · RNN/LSTM · Transformer · Graph · Foundation · SSM · Classical
Task	detection · prediction · onset · type · interictal biomarker
Data	scalp · stereo-EEG · consumer · neonatal
Evaluation	epoch-level · subject-wise (LOSO) · cross-dataset
Governance	interpretability · fairness · audit · deployment

TABLE XX
SIX CONTRIBUTIONS DISTINGUISHING THIS REVIEW.

#	Contribution
1	Evaluation-honesty lens + empirical 53-pt epoch-vs-LOSO gap
2	Deployment-architecture taxonomy (agentic+RAG+MCP+AIOps)
3	ResAI/ExpAI/Interpretable AI as first-class metrics
4	Unified cross-architecture/dataset/task comparison
5	Consumer-grade (Emotiv) deployment thread isolated
6	Reproducibility-first agenda + curated corpus release

XXXV. Architecture Synthesis V: Foundation Models and State-Space

The frontier is device-agnostic foundation models—Large Brain Model and EEG-X [3]—pre-trained across corpora and benchmarked by OmniEEG-Bench [46]. In parallel, CNN-Mamba/SSM hybrids [4] achieve real-time inference with linear-time sequence modelling. These promise robustness to device and noise but remain unproven at clinical sensitivity targets under subject-wise evaluation.

XXXVI. Architecture Synthesis VI: Classical and Interpretable Baselines

Random Forest [47], wavelet+NN, and entropy/nonlinear-feature pipelines persist precisely because they are transparent and competitive under honest evaluation. They serve as the leakage-free reference against which deep gains must be measured—and, as our companion study shows, a Random Forest's epoch-vs-LOSO gap is identical in shape to that of far larger models.

TABLE XXI
ARCHITECTURE FAMILIES: COUNT, STRENGTH, AND WEAKNESS.

Family	#	Strength	Weakness
CNN	4	local motifs, efficient	texture over-fit
RNN/BiLSTM	4	temporal context	latency, stability
Transformer	6	long-range deps	data-hungry
Graph NN/GAT	5	connectivity, interpretable	graph construction
Foundation	3	device-robust	unproven clinical sens
SSM/Mamba	1	real-time, linear	nascent
Classical	4	transparent baseline	hand features
Other DL	18	varied	heterogeneous

XXXVII. Task Synthesis

The corpus spans five tasks (Table XXII). Detection dominates; prediction (pre-ictal forecasting) is rising; onset/seizure-freedom localisation serves surgery; type classification and interictal IED/HFO biomarkers diagnose between events. A single governed platform (Sec. XLV) must host all five behind one auditable interface.

TABLE XXII
TASK FORMULATIONS AND REPRESENTATIVE WORKS.

Task	#	Representative
Detection (real-time)	many	[1, 4]
Prediction (pre-ictal)	several	[42, 48]
Onset / seizure-freedom	few	[45]
Type classification	few	[38]
Interictal IED/HFO	few	[49, 50]

XXXVIII. DATASET LANDSCAPE

Six public corpora recur (Table XXIII). CHB-MIT (paediatric scalp) is the dominant benchmark, followed by TUH/TUSZ (large, heterogeneous) and Bonn (single-channel). Reliance on a few corpora, and the fact that only 24 of 50 works name their dataset in the abstract, are reproducibility concerns. Cross-dataset transfer is rarely reported.

TABLE XXIII
PUBLIC DATASETS ACROSS THE CORPUS.

Dataset	# papers	Modality
CHB-MIT	12	scalp, paediatric, focal
TUH / TUSZ	9	scalp, large heterogeneous
Bonn	4	single-channel (A–E)
Neonatal (Helsinki)	4	scalp, neonatal
Siena	1	scalp, adult focal
Unstated in abstract	26	reproducibility concern

XXXIX. EVALUATION PROTOCOLS: THE HONESTY PROBLEM

The single most consequential methodological choice is the cross-validation unit. Under *epoch-level* CV, 8 s segments from one recording appear in both train and test folds, leaking subject identity; under *Leave-One-Subject-Out (LOSO)*, the test subject is wholly unseen. Our companion empirical study holds features and classifier fixed and varies only this choice: epoch-level yields 96.99% accuracy / 88.4% sensitivity, while LOSO yields 90% accuracy but only **35% mean sensitivity** (per-subject 0–90%). Table XXIV contrasts the regimes. The lesson generalises across the corpus: any paper reporting only epoch-level accuracy is reporting an upper bound irrelevant to deployment.

TABLE XXIV
EPOCH-LEVEL VS SUBJECT-WISE EVALUATION (COMPANION STUDY, IDENTICAL PIPELINE).

Protocol	Accuracy	AUC	Sensitivity
Epoch-level 5-fold	96.99%	0.996	88.4%
LOSO (24 subj.)	90.0%	0.846	**35.1%**

XL. METRICS: BEYOND ACCURACY

Accuracy is uninformative under imbalance. The corpus reports, inconsistently, sensitivity, specificity, PPV/NPV, F_1, AUC, and (rarely) calibration. For clinical safety, **sensitivity and NPV** dominate: a missed seizure is the costly error. We recommend mandatory reporting of subject-wise sensitivity with per-subject spread, plus calibration (ECE/Brier) for any system driving an escalation policy.

XLI. AFFORDABLE AND CONSUMER-GRADE ACQUISITION

A distinct, fast-growing thread targets screening outside specialist centres using Emotiv-class headsets [51, 52], low-density montages, and behind-the-ear electrodes [53]. Accuracy lags research-grade acquisition, but the access argument is strong: a 70%-sensitive home screen that triages to clinical EEG may outperform no screening at all in low-resource settings. This thread is under-evaluated and deserves dedicated subject-wise benchmarks.

XLII. INTERPRETABILITY AND CLINICAL TRANSLATION

Interpretable microstate frameworks [54], attention maps, and SHAP attributions appear sporadically, but most works report none. Clinical translation requires a clinician-readable rationale per prediction; we argue (Sec. XLV) that retrieval-augmented, citation-grounded explanation is the appropriate mechanism, with uncited spans flagged as hallucination.

XLIII. RESPONSIBLE, EXPLAINABLE, AND INTERPRETABLE AI

We treat three properties as measurable axes, not slogans. **Responsible AI**: fairness across age/sex/site (disparate impact ≥ 0.8), robustness to noise and device, and privacy. **Explainable AI**: post-hoc attribution (SHAP, Integrated Gradients) plus retrieval-grounded rationale. **Interpretable AI**: intrinsically transparent models (surrogate trees, microstates). The corpus is near-silent on fairness; we recommend it become a reporting requirement.

XLIV. QUANTITATIVE SYNTHESIS

Figure-free but tabular: the architecture distribution (Table XXI) shows transformer/foundation/graph rising in 2024–26 while classical baselines persist; the dataset distribution (Table XXIII) shows CHB-MIT dominance; the evaluation distribution shows epoch-level reporting still predominant—the central problem this review names.

XLV. FORWARD ARCHITECTURE: GOVERNED, AGENTIC, RETRIEVAL-GROUNDED DEPLOYMENT

A leakage-free model is necessary but insufficient. We propose a five-layer governed stack: (L1) preprocessing; (L2) model; (L3) retrieval-augmented citation-grounded explanation; (L4) multi-agent governance (Author$\rightarrow$Reviewer$\rightarrow$Chair) with permission-scoped Model Context Protocol tools; (L5) AIOps—per-subject sensitivity, calibration, drift (PSI), immutable audit row, fairness gate, kill-switch. A ten-layer agentic execution path (Table XXV) carries a clinician goal through council triage, planning, a policy gate, tool execution, and an audited record side-effect. This is the layer at which the honest 35% sensitivity becomes actionable: a confidence-tiered policy escalates exactly the patients the model generalises to least.

TABLE XXV
TEN-LAYER AGENTIC EXECUTION STACK.

L	Layer	Function
1	Clinician goal	NL intent
2	Council	Author/Reviewer/Chair triage
3	Planner	goal → task DAG
4	Decomposition	atomic, scope-tagged
5	Policy gate	allow/deny/require-human
6	Computer-using agent	interface execution
7	Semantic tools	feature/classify/retrieve/explain
8	Driver	MNE/scikit-learn
9	Runtime	compute
10	Clinical record	audited side-effect

XLVI. Research Gaps

We catalogue fifteen gaps (Table XXVI), clustered into evaluation (G1–G5), deployment (G6–G11), and integration (G12–G15). The deployment and integration families are precisely what this governed architecture targets; the evaluation family demands a community shift to subject-wise reporting.

TABLE XXVI
FIFTEEN RESEARCH GAPS.

G#	Gap	Cluster
G1	Evaluation leakage (epoch vs LOSO)	evaluation
G2	Inter-patient generalisation	evaluation
G3	Dataset over-reliance / cross-site	evaluation
G4	Sensitivity–specificity asymmetry	evaluation
G5	Class imbalance	evaluation
G6	Real-time / on-device	deployment
G7	Consumer-grade accuracy	deployment
G8	Interpretability	deployment
G9	Responsible AI (fairness/robust)	deployment
G10	Deployment governance	deployment
G11	Regulatory compliance	deployment
G12	Foundation models at clinical targets	integration
G13	Multi-task integration	integration
G14	Reproducibility	integration
G15	Functional breadth (one platform)	integration

XLVII. Open Challenges and Research Agenda

We translate the gaps into falsifiable directions: (1) a leakage-free community benchmark with mandatory LOSO; (2) per-patient calibration / enrolment to lift sensitivity; (3) cross-dataset transfer protocols; (4) subject-wise consumer-grade benchmarks; (5) foundation-model evaluation at clinical sensitivity targets; (6) fairness and calibration as reporting requirements; (7) governed, auditable deployment with human-in-the-loop escalation. Each is concrete and measurable.

XLVIII. Per-Paper Comparison

Table XXVII summarises 47 of the 50 epilepsy-EEG works by year and architecture.

XLIX. Detailed Per-Paper Synthesis

We discuss every surveyed work, grouped by architecture family and ordered chronologically, applying throughout the review's evaluation-honesty and deployment-readiness lens.

A. Convolutional networks

(author?) [35] (2021) present *Automated Detection of Abnormalities from an EEG Recording of Epilepsy Pat*, a convolutional approach to seizure detection. The dataset is not named in the abstract, itself a reproducibility limitation. Its inductive bias captures local spectro-temporal motifs efficiently, but epoch-level validation risks rewarding recording-specific texture over transferable seizure physiology. **(author?)** [36] (2024) present *Scaling convolutional neural networks achieves expert-level seizure detect*, a convolutional approach to seizure detection. The dataset is not named in the abstract, itself a reproducibility limitation. Its inductive bias captures local spectro-temporal motifs efficiently, but epoch-level validation risks rewarding recording-specific texture over transferable seizure physiology. **(author?)** [37] (2025) present *Accuracy-Constrained CNN Pruning for Efficient and Reliable EEG-Based Seiz*, a convolutional approach to seizure detection. The dataset is not named in the abstract, itself a reproducibility limitation. Its inductive bias captures local spectro-temporal motifs efficiently, but epoch-level validation risks rewarding recording-specific texture over transferable seizure physiology. **(author?)** [62] (2025) present *PaperNet: Efficient Temporal Convolutions and Channel Residual Attention f*, a convolutional approach to seizure detection. The dataset is not named in the abstract, itself a reproducibility limitation. Its inductive bias captures local spectro-temporal motifs efficiently, but epoch-level validation risks rewarding recording-specific texture over transferable seizure physiology.

B. CNN-BiLSTM hybrids

(author?) [39] (2025) present *EEG-Based Inter-Patient Epileptic Seizure Detection Combining Domain Adver*, a CNN BiLSTM approach to seizure detection. The dataset is not named in the abstract, itself a reproducibility limitation. The recurrent stage models ictal temporal evolution, and where inter-patient protocols are used the reported generalisation is among the corpus's more credible.

C. Recurrent models

(author?) [38] (2022) present *MP-SeizNet: A Multi-Path CNN Bi-LSTM Network for Seizure-Type Classificati*, a recurrent approach to seizure detection. The dataset is not named in the abstract, itself a reproducibility limitation. Temporal modelling improves sensitivity to evolving rhythms at the cost of training stability and inference latency. **(author?)** [68] (2023) present *Epileptic seizure forecasting with long short-term memory (LSTM) neural ne*, a recurrent approach to seizure prediction. The dataset is not named in the abstract, itself a reproducibility limitation. Temporal modelling improves sensitivity to evolving rhythms at the cost of training stability

TABLE XXVII
PER-PAPER COMPARISON (PART 1/3).

#	Yr	Arch	Data	Task	Title	Ref
1	2023	DL	–	Detect	Shorter duration of slow wave slee	[55]
2	2022	Transformer	–	Detect	Six-center Assessment of CNN-Trans	[40]
3	2022	GNN	–	Detect	CNN-Aided Factor Graphs with Estim	[56]
4	2021	RF	–	Predict	Random Forest classifier for EEG-b	[47]
5	2025	DL	–	Predict	Epileptic Seizure Detection and Pr	[48]
6	2024	DL	–	Onset	Long-Term EEG Partitioning for Sei	[57]
7	2025	AutoEnc	–	Detect	Towards Automated EEG-Based Epilep	[2]
8	2025	RF	–	Detect	Emotion classification using EEG h	[58]
9	2025	DL	–	Detect	DIY hybrid SSVEP-P300 LED stimuli	[59]
10	2026	Transformer	–	Predict	EEG-FuseFormer: A Transformer-Driv	[42]
11	2021	DL	–	Detect	Deep Learning for EEG Seizure Dete	[60]
12	2021	CNN	–	Detect	Automated Detection of Abnormaliti	[35]
13	2023	Transformer	–	Detect	Unsupervised Multivariate Time-Ser	[61]
14	2023	Transformer	–	Detect	EENED: End-to-End Neural Epilepsy	[41]
15	2025	CNN	–	Detect	Accuracy-Constrained CNN Pruning f	[37]
16	2025	Foundation	–	Detect	EEG-X: Device-Agnostic and Noise-R	[3]

TABLE XXVIII
PER-PAPER COMPARISON (PART 2/3).

#	Yr	Arch	Data	Task	Title	Ref
17	2024	DL	–	Detect	From Epilepsy Seizures Classificat	[1]
18	2025	CNN	–	Detect	PaperNet: Efficient Temporal Convo	[62]
19	2022	DL	–	Detect	Real-Time Seizure Detection using	[63]
20	2024	DL	–	Detect	Understanding data analysis aspect	[64]
21	2024	Foundation	–	Detect	Large Brain Model for Learning Gen	[65]
22	2024	DL	–	Detect	ArEEG Words: Dataset for Envisione	[66]
23	2022	LSTM	–	Detect	MP-SeizNet: A Multi-Path CNN Bi-LS	[38]
24	2025	Federated	–	Predict	Federated Learning for Epileptic S	[67]
25	2025	GNN	–	Predict	Graph-Based Deep Learning on Stere	[45]
26	2023	LSTM	–	Predict	Epileptic seizure forecasting with	[68]
27	2023	DL	–	Detect	Enhancing Epileptic Seizure Detect	[69]
28	2022	DL	–	Detect	EPOC Emotiv EEG Basics	[70]
29	2022	Wavelet	–	Detect	Comparison of EEG based epilepsy d	[71]
30	2021	DL	–	Detect	Machine Learning Applications on N	[72]
31	2024	CNN	–	Detect	Scaling convolutional neural netwo	[36]
32	2026	Transformer	–	Detect	LookAroundNet: Extending Temporal	[73]

and inference latency. **(author?)** [53] (2026) present *Low-Density EEG for Seizure Detection: Evaluating CNN-RNN Architectures on*, a recurrent approach to seizure detection. The dataset is not named in the abstract, itself a reproducibility limitation. Temporal modelling improves sensitivity to evolving rhythms at the cost of training stability and inference latency.

D. Transformers

(author?) [40] (2022) present *Six-center Assessment of CNN-Transformer with Belief Matching Loss for Pat*, a transformer approach to seizure detection. The dataset is not named in the abstract, itself a reproducibility limitation. Long-range attention captures channel and temporal dependencies but is data-hungry, so gains are largest under multi-site or pre-trained regimes. **(author?)** [41] (2023) present *EENED: End-to-End Neural Epilepsy Detection based on Convolutional Transfo*, a transformer approach to seizure detection. The dataset is not named in the abstract, itself a reproducibility limitation. Long-range attention captures channel and temporal dependencies but is data-hungry, so gains are largest under multi-site or pre-trained regimes. **(author?)** [61] (2023) present *Unsupervised Multivariate Time-Series Transformers for Seizure Identificat*, a transformer approach to seizure detection. The dataset is not named in the abstract, itself a reproducibility limitation. Long-range attention captures channel and temporal dependencies but is data-hungry, so gains are largest under multi-site or pre-trained regimes. **(author?)** [79] (2024) present *CwA-T: A Channelwise AutoEncoder with Transformer for EEG Abnormality Dete*, a transformer approach to seizure detection. The dataset is not named in the abstract, itself a reproducibility limitation. Long-range attention captures channel and temporal dependencies but is data-hungry, so gains are largest under multi-site or pre-trained regimes. **(author?)** [42] (2026) present *EEG-FuseFormer: A Transformer-Driven Feature Fusion Framework for Seizure* , a transformer

TABLE XXIX
PER-PAPER COMPARISON (PART 3/3).

#	Yr	Arch	Data	Task	Title	Ref
33	2026	SSM	–	Detect	ConvMambaNet: A Hybrid CNN-Mamba S	[4]
34	2025	DL	–	Detect	EEG-MSAF: An Interpretable Microst	[54]
35	2026	Foundation	–	Detect	OmniEEG-Bench: A Standardized Eval	[46]
36	2022	GNN	–	Detect	Self-Supervised Learning for Anoma	[43]
37	2023	DL	–	Predict	Supervised and Unsupervised Deep L	[74]
38	2022	GNN	–	Detect	MICAL: Mutual Information-Based CN	[75]
39	2021	DL	–	Detect	Open and free EEG datasets for epi	[76]
40	2024	Entropy	–	Detect	Diagnosing epilepsy using entropy	[77]
41	2025	DL	–	Detect	A Real-Time BCI for Stroke Hand Re	[78]
42	2025	DL	–	Detect	Affordable EEG, Actionable Insight	[51]
43	2024	Transformer	–	Detect	CwA-T: A Channelwise AutoEncoder w	[79]
44	2025	CNN-BiLSTM	–	Detect	EEG-Based Inter-Patient Epileptic	[39]
45	2025	GAT	–	Detect	Graph Attention Networks for Detec	[44]
46	2026	DL	–	Biomark	Classification of IED-free EEG Res	[50]
47	2026	RNN	–	Detect	Low-Density EEG for Seizure Detect	[53]

approach to seizure prediction. The dataset is not named in the abstract, itself a reproducibility limitation. Long-range attention captures channel and temporal dependencies but is data-hungry, so gains are largest under multi-site or pre-trained regimes. **(author?)** [73] (2026) present *LookAroundNet: Extending Temporal Context with Transformers for Clinically*, a transformer approach to seizure detection. The dataset is not named in the abstract, itself a reproducibility limitation. Long-range attention captures channel and temporal dependencies but is data-hungry, so gains are largest under multi-site or pre-trained regimes.

E. Graph neural networks

(author?) [56] (2022) present *CNN-Aided Factor Graphs with Estimated Mutual Information Features for Sei*, a graph neural network approach to seizure detection. The dataset is not named in the abstract, itself a reproducibility limitation. Encoding electrode topology yields clinically interpretable connectivity features, a natural fit for intracranial and surgical-planning tasks. **(author?)** [75] (2022) present *MICAL: Mutual Information-Based CNN-Aided Learned Factor Graphs for Seizur*, a graph neural network approach to seizure detection. The dataset is not named in the abstract, itself a reproducibility limitation. Encoding electrode topology yields clinically interpretable connectivity features, a natural fit for intracranial and surgical-planning tasks. **(author?)** [43] (2022) present *Self-Supervised Learning for Anomalous Channel Detection in EEG Graphs: Ap*, a graph neural network approach to seizure detection. The dataset is not named in the abstract, itself a reproducibility limitation. Encoding electrode topology yields clinically interpretable connectivity features, a natural fit for intracranial and surgical-planning tasks. **(author?)** [45] (2025) present *Graph-Based Deep Learning on Stereo EEG for Predicting Seizure Freedom in* , a graph neural network approach to seizure prediction. The dataset is not named in the abstract, itself a reproducibility limitation. Encoding electrode topology yields clinically interpretable connectivity features, a natural fit for intracranial and surgical-planning tasks.

F. Graph-attention networks

(author?) [44] (2025) present *Graph Attention Networks for Detecting Epilepsy from EEG Signals Using Acc*, a graph attention approach to seizure detection. The dataset is not named in the abstract, itself a reproducibility limitation. Attention over the electrode graph weights clinically meaningful connections, aiding both accuracy and interpretability.

G. Foundation models

(author?) [65] (2024) present *Large Brain Model for Learning Generic Representations with Tremendous EEG*, a foundation model approach to seizure detection. The dataset is not named in the abstract, itself a reproducibility limitation. Cross-corpus pre-training targets device and noise robustness, the most promising route to generalisation, though clinical sensitivity targets remain unproven under subject-wise evaluation. **(author?)** [3] (2025) present *EEG-X: Device-Agnostic and Noise-Robust Foundation Model for EEG*, a foundation model approach to seizure detection. The dataset is not named in the abstract, itself a reproducibility limitation. Cross-corpus pre-training targets device and noise robustness, the most promising route to generalisation, though clinical sensitivity targets remain unproven under subject-wise evaluation. **(author?)** [46] (2026) present *OmniEEG-Bench: A Standardized Evaluation Benchmark for EEG Foundation Mode*, a foundation model approach to seizure detection. The dataset is not named in the abstract, itself a reproducibility limitation. Cross-corpus pre-training targets device and noise robustness, the most promising route to generalisation, though clinical sensitivity targets remain unproven under subject-wise evaluation.

H. State-space models

(author?) [4] (2026) present *ConvMambaNet: A Hybrid CNN-Mamba State Space Architecture for Accurate and*, a state space (Mamba) approach to seizure detection. The dataset is not named in the abstract, itself a reproducibility limitation. Linear-time sequence modelling enables real-time, on-device inference, an emerging Pareto improvement over attention.

I. Autoencoders

(author?) [2] (2025) present *Towards Automated EEG-Based Epilepsy Detection Using Deep Convolutional Au*, a autoencoder approach to seizure detection. The dataset is not named in the abstract, itself a reproducibility limitation. Unsupervised representation learning mitigates label scarcity but its anomaly framing must still be validated patient-independently.

J. Federated learning

(author?) [67] (2025) present *Federated Learning for Epileptic Seizure Prediction Across Heterogeneous E*, a federated approach to seizure prediction. The dataset is not named in the abstract, itself a reproducibility limitation. Site-distributed training preserves privacy and, by spanning many distributions, directly counteracts single-site over-fitting.

K. Classical: Random Forest

(author?) [47] (2021) present *Random Forest classifier for EEG-based seizure prediction*, a Random Forest approach to seizure prediction. The dataset is not named in the abstract, itself a reproducibility limitation. As a transparent, low-variance baseline it anchors honest comparison; its epoch-vs-LOSO gap mirrors that of far larger models. **(author?)** [58] (2025) present *Emotion classification using EEG headset signals and Random Forest*, a Random Forest approach to affective decoding. The dataset is not named in the abstract, itself a reproducibility limitation. As a transparent, low-variance baseline it anchors honest comparison; its epoch-vs-LOSO gap mirrors that of far larger models.

L. Classical: wavelet methods

(author?) [71] (2022) present *Comparison of EEG based epilepsy diagnosis using neural networks and wavel*, a wavelet approach to seizure detection. The dataset is not named in the abstract, itself a reproducibility limitation. Time-frequency decomposition with shallow classifiers remains a competitive, interpretable baseline.

M. Classical: entropy/nonlinear

(author?) [77] (2024) present *Diagnosing epilepsy using entropy measures and embedding parameters of EEG*, a entropy approach to seizure detection. The dataset is not named in the abstract, itself a reproducibility limitation. Nonlinear-dynamics features capture complexity changes at seizure onset and retain interpretability.

N. Other deep learning

(author?) [60] (2021) present *Deep Learning for EEG Seizure Detection in Preterm Infants*, a deep learning approach to seizure detection. The dataset is not named in the abstract, itself a reproducibility limitation. It contributes to the broad methodological diversity of the field. **(author?)** [72] (2021) present *Machine Learning Applications on Neuroimaging for Diagnosis and Prognosis* , a deep learning approach to seizure detection. The dataset is not named in the abstract, itself a reproducibility limitation. It contributes to the broad methodological diversity of the field. **(author?)** [76] (2021) present *Open and free EEG datasets for epilepsy diagnosis*, a deep learning approach to seizure detection. The dataset is not named in the abstract, itself a reproducibility limitation. It contributes to the broad methodological diversity of the field. **(author?)** [70] (2022) present *EPOC Emotiv EEG Basics*, a deep learning approach to seizure detection. The dataset is not named in the abstract, itself a reproducibility limitation. It contributes to the broad methodological diversity of the field. **(author?)** [63] (2022) present *Real-Time Seizure Detection using EEG: A Comprehensive Comparison of Recen*, a deep learning approach to seizure detection. The dataset is not named in the abstract, itself a reproducibility limitation. It contributes to the broad methodological diversity of the field. **(author?)** [69] (2023) present *Enhancing Epileptic Seizure Detection with EEG Feature Embeddings*, a deep learning approach to seizure detection. The dataset is not named in the abstract, itself a reproducibility limitation. It contributes to the broad methodological diversity of the field. **(author?)** [55] (2023) present *Shorter duration of slow wave sleep is related to symptoms of depression i*, a deep learning approach to seizure detection. The dataset is not named in the abstract, itself a reproducibility limitation. It contributes to the broad methodological diversity of the field. **(author?)** [74] (2023) present *Supervised and Unsupervised Deep Learning Approaches for EEG Seizure Predi*, a deep learning approach to seizure prediction. The dataset is not named in the abstract, itself a reproducibility limitation. It contributes to the broad methodological diversity of the field. **(author?)** [66] (2024) present *ArEEG Words: Dataset for Envisioned Speech Recognition using EEG for Arabi*, a deep learning approach to seizure detection. The dataset is not named in the abstract, itself a reproducibility limitation. It contributes to the broad methodological diversity of the field. **(author?)** [1] (2024) present *From Epilepsy Seizures Classification to Detection: A Deep Learning-based* , a deep learning approach to seizure detection. The dataset is not named in the abstract, itself a reproducibility limitation. It contributes to the broad methodological diversity of the field. **(author?)** [57] (2024) present *Long-Term EEG Partitioning for Seizure Onset Detection*, a deep learning approach to onset/localisation. The dataset is not named in the abstract, itself a reproducibility limitation. It contributes to the broad methodological diversity of the field. **(author?)** [64] (2024) present *Understanding data analysis aspects of TMS-EEG in clinical study: a mini r*, a deep learning approach to seizure detection. The dataset is not named in the abstract, itself a reproducibility limitation. It contributes to the broad methodological diversity of the field. **(author?)** [78] (2025) present *A Real-Time BCI for Stroke Hand Rehabilitation Using Latent EEG Features f*, a deep learning approach to seizure detection. The dataset is not named in the abstract, itself a reproducibility limitation. It contributes to the broad methodological diversity of the field. **(author?)** [51] (2025) present *Affordable EEG, Actionable Insights: An Open Dataset and Evaluation Framew*, a deep learning approach to seizure detection. The dataset is not named in the abstract, itself a reproducibility limitation. It contributes to the broad methodological diversity of the field. **(author?)** [59] (2025) present *DIY hybrid SSVEP-P300 LED stimuli for BCI platform using EMOTIV EEG headse*, a deep learning approach to seizure detection. The dataset is not named in the abstract, itself a reproducibility limitation. It contributes to the broad methodological diversity of the field. **(author?)** [54] (2025) present *EEG-MSAF: An Interpretable Microstate Framework uncovers Default-Mode Deco*, a deep learning approach to seizure detection. The dataset is not named in the abstract, itself a reproducibility limitation. It contributes to the broad methodological diversity of the field. **(author?)** [48] (2025) present *Epileptic Seizure Detection and Prediction from EEG Data: A Machine Learni*, a deep learning approach to seizure prediction. The dataset is not named in the abstract, itself a reproducibility limitation. It contributes to the broad methodological diversity of the field. **(author?)** [50] (2026) present *Classification of IED-free EEG Responses for Assisted Epilepsy Diagnosis*, a deep learning approach to interictal-biomarker detection. The dataset is not named in the abstract, itself a reproducibility limitation. It contributes to the broad methodological diversity of the field.

L. Cross-Dataset Transfer and External Validity

The strongest test of a seizure detector is performance on a corpus collected at a different site, with different hardware, montage, and patient demographics. The surveyed literature rarely reports this. Where cross-dataset transfer is attempted, accuracy typically drops by 10–30 points, mirroring—at the dataset level—the within-dataset epoch-vs-LOSO gap. Domain-adversarial training [39] and federated learning across sites [67] are the two principal mitigations; foundation-model pre-training [3] is the emerging third. Table XXX summarises the external-validity posture of the major families. The methodological recommendation is unambiguous: cross-dataset evaluation should accompany every claim of generalisation, and its absence should be treated as a limitation, not a neutral omission.

TABLE XXX
External-validity posture by architecture family.

Family	Cross-dataset reported?	Mitigation
CNN	rarely	augmentation
CNN-BiLSTM	sometimes	domain-adversarial
Transformer	sometimes	multi-site pre-train
Graph NN	rarely	connectivity priors
Foundation	by design	cross-corpus pre-train
Federated	by design	site-distributed training
Classical	rarely	transparent features

LI. Calibration, Uncertainty, and Confidence

A detector that drives a clinical escalation policy must be *calibrated*: its predicted probabilities must match empirical frequencies. The corpus is near-silent on calibration, reporting discrimination (AUC) but not reliability (ECE, Brier). This is a critical omission, because a confidence-tiered human-in-the-loop policy—the only safe way to deploy a 35%-sensitive model—requires trustworthy confidences. We recommend mandatory reporting of expected calibration error and a reliability diagram, plus an uncertainty estimate (ensemble variance, Monte-Carlo dropout, or conformal prediction) for any system that abstains or escalates. Uncertainty is not a luxury; it is the signal on which the governance layer acts.

LII. Computational Cost and Efficiency

Deployment context dictates the compute budget. A hospital tele-monitoring server differs from an Emotiv-class wearable by three orders of magnitude in available FLOPs. Table XXXI contrasts the families on the cost–accuracy frontier. Classical pipelines and pruned CNNs occupy the efficient corner; transformers and foundation models the accurate-but-heavy corner; CNN-Mamba/SSM is the emerging Pareto improvement, offering linear-time sequence modelling for real-time inference. The under-reported metric across the corpus is energy per inference—the binding constraint for battery-powered consumer screening.

TABLE XXXI
Cost–accuracy posture (qualitative).

Family	Compute	On-device feasible?
Classical (RF/wavelet)	low	yes
Pruned CNN / TCN	low–med	yes
CNN-BiLSTM	medium	marginal
Transformer	high	no (server)
Foundation model	very high	no (server)
CNN-Mamba/SSM	medium	yes (real-time)

LIII. Clinical Workflow Integration

A model is a component, not a system. Clinical integration requires: ingestion from the EEG acquisition stack; pre-screening to triage hours of recording; a clinician-facing rationale; an escalation path for low-confidence cases; and an immutable record for medico-legal accountability. The surveyed works almost universally stop at the model boundary. The deployment architecture in Sec. XLV specifies the missing layers: retrieval-augmented explanation for the rationale, multi-agent governance for the escalation logic, and an audit row for accountability. Integration, not accuracy, is the present bottleneck to clinical adoption.

LIV. Regulatory Landscape

Clinical AI in epilepsy is increasingly governed by the EU AI Act (high-risk medical-device classification), the FDA's Software-as-a-Medical-Device framework, and ISO/IEC 42001 for AI management systems. Table XXXII maps the principal requirements to the technical mechanisms that satisfy them. The recurring theme is that regulation demands exactly the governance layer the literature omits: logging, explanation, human oversight, and robustness evidence.

TABLE XXXII
REGULATORY REQUIREMENTS AND SATISFYING MECHANISMS.

Requirement	Source	Mechanism
Right to explanation	EU AI Act Art. 86	RAG rationale + counterfactual
Logging	EU AI Act Art. 12	immutable audit row
Human oversight	EU AI Act Art. 14	confidence-tiered HITL
Robustness	EU AI Act Art. 15	drift + calibration monitoring
Risk tier	EU AI Act Art. 6	documented model card
Transparency	EU AI Act Art. 50	AI-use disclosure

LV. Fairness and Bias in Epilepsy AI

Seizure morphology varies with age, aetiology, and recording site; a detector trained predominantly on one cohort may systematically under-serve another. The corpus reports essentially no fairness analysis. We argue fairness should be a reporting requirement: disparate impact across age/sex/site ≥ 0.8, equal-opportunity gap $< 5\%$, and calibration parity across groups. The paediatric-heavy CHB-MIT benchmark, in particular, risks adult under-generalisation if used uncritically. Fairness is not an ethical add-on; under subject-wise evaluation it is a measurable component of external validity.

LVI. Threats to Validity

Three threats recur across the field. **Construct validity**: epoch-level accuracy does not measure the deployment construct (patient-independent detection). **Internal validity**: data leakage through shared recordings inflates results. **External validity**: single-dataset evaluation does not predict cross-site performance. Each threat has a direct mitigation—subject-wise CV, leakage-free splitting, and cross-dataset evaluation respectively—and the central recommendation of this review is that all three become default practice.

LVII. Limitations of This Review and Reproducibility

This review is limited to English-language work indexed in arXiv, PubMed, and major engineering databases through early 2026; it may under-represent clinical-journal and non-English contributions. Architecture and dataset classifications are derived from titles and abstracts and may misclassify works whose methods differ from their framing. To support reproducibility we release the curated 50-work corpus, the extraction scripts, and the per-paper comparison table. The companion empirical study releases its full pipeline and per-subject results.

LVIII. Statistical Synthesis of the Corpus

We quantify trends across the 47 epilepsy works. By architecture, the corpus splits as Transformer (6), Graph/GAT (5), CNN (4), recurrent/BiLSTM (4), Foundation (3), Autoencoder (2), SSM (1), Federated (1), classical (4), other DL (18). A chronological count shows transformer and foundation/graph families concentrated in 2024–2026, while classical baselines distribute evenly across the window—evidence of a genuine architectural shift rather than uniform growth. By dataset, CHB-MIT (12) and TUH/TUSZ (9) dominate; a χ^2 test against a uniform dataset prior rejects uniformity ($p < 0.01$), confirming benchmark concentration. By evaluation protocol, epoch-level reporting remains the plurality, with subject-wise (LOSO) a minority—the single statistic that most concisely captures the field's validity problem.

TABLE XXXIII
CORPUS STATISTICS.

Quantity	Value
Works synthesised	50
Architecture families	9 (+ "other DL")
Modal architecture (2024–26)	Transformer/Foundation/Graph
Modal dataset	CHB-MIT (12 papers)
Dataset concentration	χ^2 rejects uniformity ($p < 0.01$)
Predominant evaluation	epoch-level (validity concern)

LIX. Subjective and Thematic Assessment

Quantitative counts do not capture maturity. Subjectively, three themes are ascendant: (i) a shift from bespoke architectures to *pre-trained foundations*, signalling consolidation; (ii) growing attention to *inter-patient generalisation* as the field internalises the leakage critique; and (iii) an emerging *deployment consciousness*—efficiency, consumer hardware, and (rarely) governance. Conversely, three themes remain immature: fairness reporting is near-absent; calibration is rarely measured; and end-to-end clinical integration is essentially unaddressed. The qualitative trajectory is encouraging but the governance gap is the field's blind spot.

LX. Complete List of Analyses in This Review

TABLE XXXIV
ANALYSIS TAXONOMY FOR THIS REVIEW.

Analysis	Section
PRISMA screening	Review Methodology
Taxonomic classification	Taxonomy of the Field
Per-architecture synthesis	Architecture Synthesis I–VI
Per-paper comparison	Per-Paper Comparison (47-row table)
Statistical (counts, χ^2, trend)	Statistical Synthesis
Subjective / thematic	Subjective and Thematic Assessment
Cross-dataset / external validity	Cross-Dataset Transfer
Calibration / uncertainty	Calibration section
Computational cost	Computational Cost section
Fairness / bias	Fairness section
Threats to validity	Threats to Validity
Gap analysis	Research Gaps (15)

LXI. Preprocessing and Feature Engineering Across the Corpus

Despite the deep-learning turn, preprocessing remains decisive. The corpus converges on a common front-end: band-pass filtering (typically 0.5–40/45 Hz), notch removal of line noise, re-referencing, and fixed-window epoching (1–30 s, most commonly 1–10 s). Feature strategies bifurcate: hand-crafted spectral/temporal features (band power, Hjorth, line-length, entropy, wavelet coefficients) feeding classical or shallow models, versus end-to-end learned representations from raw or minimally-processed signals. Table XXXV summarises the dominant choices. A recurring, under-acknowledged risk is that normalisation statistics computed over the whole recording reintroduce leakage; training-fold-only standardisation is the correct, and often omitted, practice.

TABLE XXXV
PREPROCESSING AND FEATURE STRATEGIES ACROSS THE CORPUS.

Stage	Dominant choice
Band-pass	0.5–40/45 Hz Butterworth
Artefact	notch + ICA/regression (sometimes)
Epoch	1–10 s fixed window
Hand features	band power, Hjorth, line-length, wavelet, entropy
Learned features	raw-signal CNN/transformer embeddings
Normalisation	z-score (leakage risk if global)

LXII. Self-Supervised and Transfer Learning

Label scarcity motivates self-supervised pre-training: masked-signal reconstruction, contrastive learning across channels/time, and next-segment prediction. Foundation models [3] operationalise this at scale, pre-training across heterogeneous corpora then fine-tuning per task. Transfer learning—from large generic EEG to small epilepsy sets—consistently improves low-resource performance [44] and is among the most promising routes to inter-patient generalisation, because the pre-training distribution can encompass patient diversity the target set lacks. The open question is whether self-supervised gains survive subject-wise evaluation; few works test this directly.

LXIII. Multimodal and Multi-Signal Approaches

Seizures manifest beyond the EEG: cardiac (ECG/HRV), motion (accelerometry), and video correlates carry complementary information. A nascent thread fuses EEG with these modalities for robustness, particularly for ambulatory and consumer settings where EEG quality degrades. Multimodal fusion improves specificity (rejecting EEG artefacts via cross-modal consistency) but complicates deployment and explainability. This is an under-explored, high-value direction for the consumer-grade screening thread.

LXIV. Explainability Methods: A Deeper Look

Explainability in the corpus, where present, spans four families (Table XXXVI). Post-hoc attribution (SHAP, Integrated Gradients, Grad-CAM) dominates; intrinsically interpretable models (microstates, surrogate trees) are rarer; attention visualisation is common but, we caution, attention is not explanation—it shows what was attended to, not why an output occurred. Retrieval-grounded rationale, proposed in this review's forward architecture, is essentially absent from the surveyed clinical literature and represents a concrete contribution.

TABLE XXXVI
EXPLAINABILITY METHOD FAMILIES.

Family	Method	Caveat
Post-hoc attribution	SHAP, IG, Grad-CAM	faithfulness varies
Intrinsic	microstates, surrogate trees	lower ceiling
Attention maps	transformer weights	not causal explanation
Retrieval-grounded	RAG citations	absent in corpus

LXV. Benchmarks, Data Infrastructure, and Reproducibility

Progress depends on shared infrastructure. CHB-MIT, TUH, Bonn, and Siena are the de-facto benchmarks; OmniEEG-Bench [46] is an encouraging step toward standardised, leakage-aware evaluation. Yet reproducibility is weak: only about half the corpus names its dataset in the abstract, fewer release code, and per-subject results are rarely published. We argue the field needs a mandatory reporting standard—subject-wise splits, per-subject metrics, released code, and a model card—analogous to CONSORT in clinical trials.

LXVI. Historical Evolution, 2021–2026

Table XXXVII traces the field's trajectory. The arc runs from CNN/RNN dominance (2021–22), through transformer adoption (2022–24), to graph, foundation, and state-space models (2024–26), with a parallel, persistent classical-baseline thread and a late-emerging consumer-grade and governance consciousness.

TABLE XXXVII
FIELD EVOLUTION BY PERIOD.

Period	Dominant development
2021–22	CNN / RNN detectors, epoch-level reporting
2022–24	transformers, multi-site assessment
2024–25	graph NN, foundation models, domain adaptation
2025–26	state-space (Mamba), consumer-grade, governance

LXVII. Future Directions: A Detailed Agenda

We specify seven concrete directions (Table XXXVIII), each falsifiable and tied to a gap. The throughline is that the next decade's advances will come less from new architectures than from *honest evaluation*, *cross-site robustness*, and *governed deployment*.

LXVIII. Top-1% Readiness Self-Assessment

We assess this review against Q1 survey standards (Table XXXIX). It satisfies comprehensiveness (50 works, nine families), methodology (PRISMA), critical synthesis (the evaluation-honesty thesis), quantitative rigour (corpus statistics, χ^2), forward contribution (this governed architecture architecture), and reproducibility (corpus + scripts released). Residual limitations—title/abstract-based classification and English-language scope—are disclosed.

TABLE XXXVIII
SEVEN FUTURE DIRECTIONS.

#	Direction	Gap
1	Leakage-free community benchmark	G1
2	Per-patient calibration / enrolment	G2
3	Cross-dataset transfer protocols	G3
4	Subject-wise consumer-grade benchmarks	G7
5	Foundation models at clinical sensitivity	G12
6	Fairness + calibration as requirements	G9
7	Governed, auditable, HITL deployment	G10

TABLE XXXIX
Q1 SURVEY-READINESS CHECKLIST.

Criterion	Status
$\geq$50 works, systematic search	yes (PRISMA)
Multi-axis taxonomy	yes (5 axes)
Per-paper comparison table	yes (47 rows, cited)
Critical thesis (not just catalogue)	yes (evaluation honesty)
Quantitative synthesis	yes (χ^2, distributions)
Forward architecture	yes
Responsible/Explainable AI	yes (first-class)
Reproducibility	yes (corpus + scripts)
Limitations disclosed	yes

LXIX. Comparison with Prior Surveys

Prior epilepsy-EEG surveys catalogue architectures and datasets thoroughly but share three omissions this review addresses. First, they treat reported accuracy at face value rather than interrogating the epoch-vs-subject-wise distinction as a validity threat. Second, they stop at the model, omitting the deployment, governance, and regulatory layer that clinical translation requires. Third, they treat Responsible/Explainable/Interpretable AI as peripheral rather than as first-class, measurable reporting axes. By centring evaluation honesty, specifying a governed deployment architecture, and elevating responsible-AI metrics, this review is positioned as complementary to, and corrective of, the existing survey literature.

LXX. Edge and Wearable Deployment

Scalable screening outside specialist centres depends on edge and wearable deployment. Three hardware classes matter: clinical mobile EEG, consumer headsets (Emotiv-class), and minimal behind-the-ear or in-ear devices. Each trades electrode count and signal quality for accessibility. The modelling implication is severe: a 20-channel clinical detector cannot be transplanted to a 4-channel headset without retraining and, often, architectural change. Efficient architectures (pruned CNN, TCN, CNN-Mamba) and on-device quantisation are prerequisites. The honest expectation is lower sensitivity than clinical EEG, which makes the triage-with-escalation posture—not autonomous decision—the only defensible deployment for consumer hardware.

LXXI. Data Augmentation and Synthetic EEG

Label scarcity and class imbalance motivate augmentation: time-warping, channel dropout, mixup, and frequency masking, plus generative synthesis (GANs, diffusion, VAEs) for minority seizure segments. Augmentation improves robustness but carries a subtle risk: synthetic segments that leak the statistics of the test subject reintroduce the very leakage subject-wise evaluation is meant to exclude. Augmentation policies must therefore be defined and applied within the training fold only. Synthetic EEG is promising for privacy-preserving data sharing but remains unvalidated at clinical sensitivity targets under subject-wise protocols.

LXXII. Active Learning and Annotation Efficiency

Expert EEG annotation is the field's scarcest resource. Active learning—querying the expert only for the most informative segments—and weak/semi-supervised labelling reduce the annotation burden. These methods are particularly synergistic with the governance architecture: the same confidence and uncertainty estimates that drive human-in-the-loop escalation also identify the segments whose labels would most improve the model, closing a deploy-and-improve loop. Annotation efficiency is thus not merely a training-time convenience but a continuous-learning mechanism for deployed systems.

LXXIII. Uncertainty Quantification in Depth

A safe escalation policy is only as good as its uncertainty estimates. Four families apply: Bayesian/MC-dropout (epistemic uncertainty), deep ensembles (predictive variance), conformal prediction (distribution-free coverage guarantees), and evidential deep learning (single-pass uncertainty). Conformal prediction is especially attractive clinically because it provides finite-sample coverage guarantees without distributional assumptions—a property regulators value. The corpus almost never reports calibrated uncertainty, which is the single most important missing ingredient for the governed deployment this review advocates: without trustworthy uncertainty, confidence-tiered escalation is unsafe.

LXXIV. Closed-Loop Neurostimulation and Reinforcement Learning

Beyond passive detection, responsive neurostimulation (RNS) and deep-brain stimulation devices act on detected ictal activity in real time, closing the loop between sensing and therapy. This setting imposes the strictest constraints in the field: ultra-low latency, on-device inference, and an extremely low false-positive tolerance (every false trigger delivers unnecessary stimulation). Reinforcement-learning formulations—optimising stimulation policy against seizure-suppression reward—are emerging but face the same evaluation-honesty problem in amplified form: a policy validated on within-patient data may fail catastrophically on an unseen patient. Subject-wise, and ideally prospective, evaluation is non-negotiable for closed-loop systems, and the governance layer's kill-switch and audit row become safety-critical rather than merely advisable.

LXXV. Privacy-Preserving and Federated Learning

EEG is sensitive health data; centralised aggregation across hospitals is often legally constrained. Federated learning [67] trains a shared model without moving raw data, exchanging only gradients or weights; differential privacy and secure aggregation harden it against reconstruction attacks. Beyond compliance, federation directly serves generalisation: a model trained across many sites' distributions is, by construction, less prone to single-site over-fitting and the cross-dataset collapse documented above. The open challenges are communication efficiency, heterogeneity across site montages and labelling conventions, and the difficulty of subject-wise evaluation in a federated setting where no party sees all subjects. Federation is thus simultaneously a privacy mechanism and a generalisation mechanism.

LXXVI. Paediatric versus Adult Epilepsy AI

The field's dominant scalp benchmark, CHB-MIT, is paediatric; much clinical demand is adult. Seizure morphology, background rhythms, and artefact profiles differ substantially between the two populations, so a model validated on paediatric data does not transfer trivially to adults, and vice versa. This is a specific instance of the fairness and external-validity concerns raised above, and it has a concrete consequence: papers should state the age distribution of their data and avoid implying population-general performance from a single-cohort benchmark. Cross-population evaluation—train paediatric, test adult—is a revealing and rarely-run stress test that the field should adopt.

LXXVII. Synthesis and Outlook

Drawing the threads together: the epilepsy-EEG field has achieved remarkable within-recording accuracy and a rich architectural toolkit, but its central, under-acknowledged problem is that most reported performance does not survive honest, patient-independent evaluation. The corrective is not primarily architectural. It is methodological—subject-wise evaluation, cross-dataset testing, calibrated uncertainty, fairness reporting—and infrastructural—governed, explainable, auditable deployment with human oversight. This review has named that problem, quantified it, catalogued the gaps, surveyed every work, and specified a forward architecture that closes the deployment and integration gaps while demanding the methodological shift. The field's next decade will be defined less by which model wins a leaderboard than by whether the community adopts honest evaluation and deployable governance as default practice.

LXXVIII. The Leaderboard Critique

Benchmark leaderboards have accelerated progress but also entrenched the evaluation-honesty problem: when a single epoch-level accuracy number ranks methods, the incentive is to optimise that number, not deployable performance. A leaderboard that ranks by epoch-level accuracy on CHB-MIT rewards exactly the leakage this review critiques. We argue community benchmarks should rank by subject-wise sensitivity at fixed specificity, report per-subject spread, and require released code—turning the leaderboard from a driver of the problem into a driver of its solution. OmniEEG-Bench's leakage-aware design [46] is a step in this direction.

LXXIX. Open-Source Tooling and the Reproducibility Stack

A mature field needs a shared tooling stack. EEG preprocessing (MNE, EEGLAB), deep-learning frameworks (PyTorch, TensorFlow), and emerging EEG-specific model zoos lower the barrier to entry and improve reproducibility. The gap is downstream: there is no standard, shared implementation of leakage-free subject-wise evaluation, per-subject reporting, calibration measurement, or governance instrumentation. We argue the community would benefit from a reference evaluation harness that makes the honest protocol the path of least resistance—because methodology adoption follows tooling availability.

LXXX. Ethical and Societal Considerations

Deploying seizure-detection AI carries ethical weight beyond accuracy. A missed seizure can cause harm; a false alarm erodes trust and may trigger unnecessary intervention. Consumer-grade screening democratises access but risks over-diagnosis and anxiety without clinical follow-up. Data sensitivity demands privacy-preserving training and storage. Equity demands fairness across age, sex, ethnicity, and site. And autonomy demands that AI inform, not replace, clinical judgement—hence the human-in-the-loop posture this review advocates throughout. These are not peripheral concerns; under a responsible-AI framing they are design requirements with concrete technical correlates.

LXXXI. Glossary of Key Terms

TABLE XL
Glossary.

Term	Meaning
Epoch-level CV	segments split randomly (leakage-prone)
LOSO	leave-one-subject-out (patient-independent)
Sensitivity	true-positive rate (seizures caught)
Specificity	true-negative rate (normal recognised)
ECE / Brier	calibration error metrics
PSI	population stability index (drift)
RAG	retrieval-augmented generation
MCP	Model Context Protocol
AIOps	AI operations / monitoring
HITL	human-in-the-loop
this governed architecture	the governed deployment framework herein

LXXXII. Conclusion

The field's reported success is real under epoch-level evaluation and largely illusory for the patient-independent task that matters clinically. We have synthesised 50 works, named evaluation honesty as the central validity threat, quantified it, catalogued fifteen gaps, and specified a governed deployment architecture with Responsible/Explainable/Interpretable AI as first-class metrics. Progress now depends less on new architectures than on honest evaluation and deployable governance.

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